RetailPulse - AI-Powered Customer Analytics & Demand Forecasting

Transforming Retail Intelligence with AI-Driven Insights

Zidio Internship Project

Developed By:

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Project Duration: July 2026 – August 2026 (1 Month)

Project Overview

RetailPulse is an AI-powered retail analytics platform designed to help businesses understand customer behavior, forecast product demand, and optimize inventory management using Machine Learning and MLOps practices. The project focuses on improving business intelligence through predictive analytics and real-time visual dashboards.

Vision & Objectives:

- **Predict future product demand using AI forecasting models.**
- **Analyze customer purchasing behavior.**
- **Identify churn-prone customers.**
- **Improve inventory planning and reduce overstocking.**

Target Users / Use Cases:

- **Retail store managers**
- **E-commerce businesses**
- **Supply chain teams**
- **Business analysts and decision-makers**

Business Value Delivered:

- **Better forecasting accuracy**
- **Reduced inventory wastage**
- **Improved customer retention**
- **Data-driven decision making**

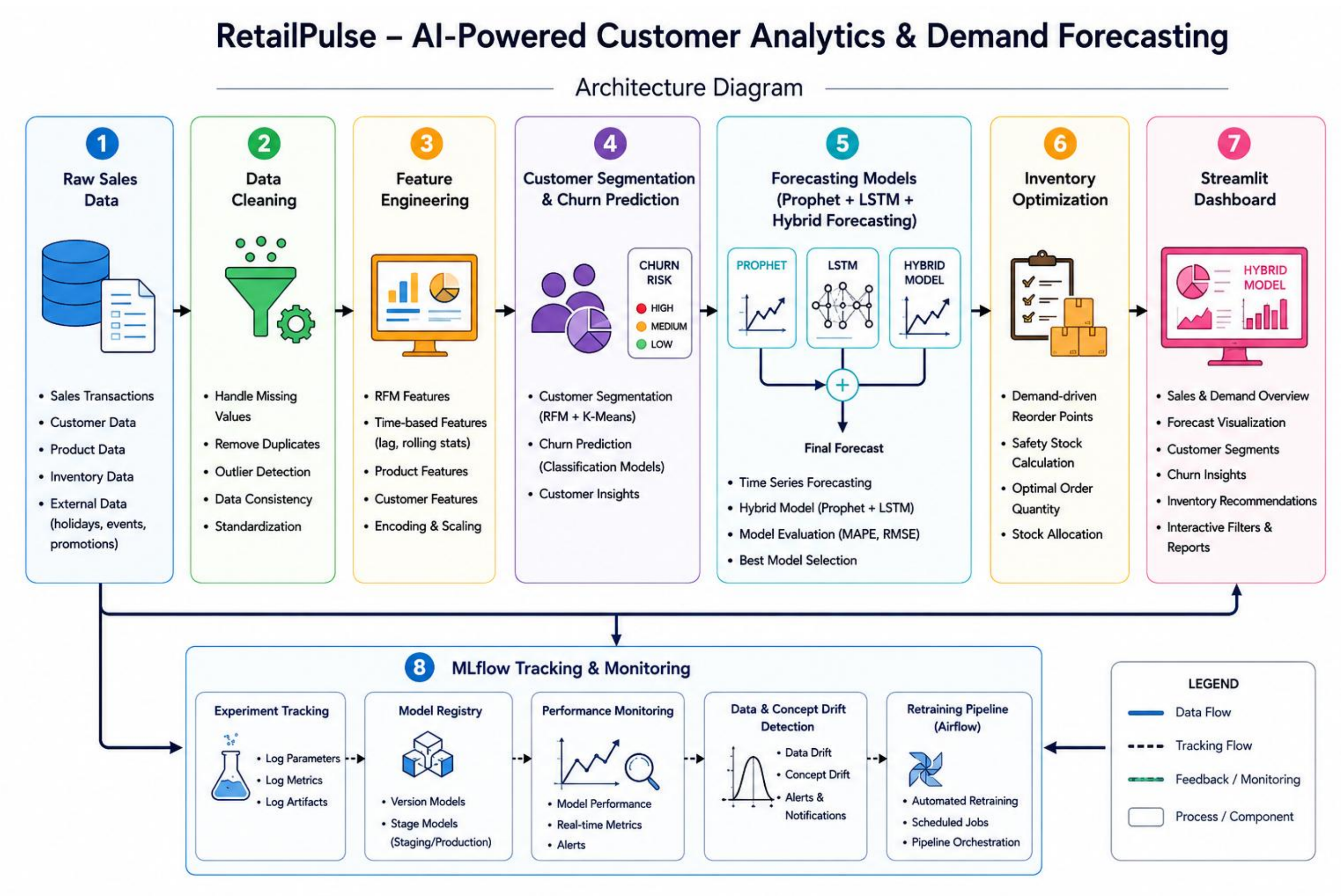
Non-functional Goals:

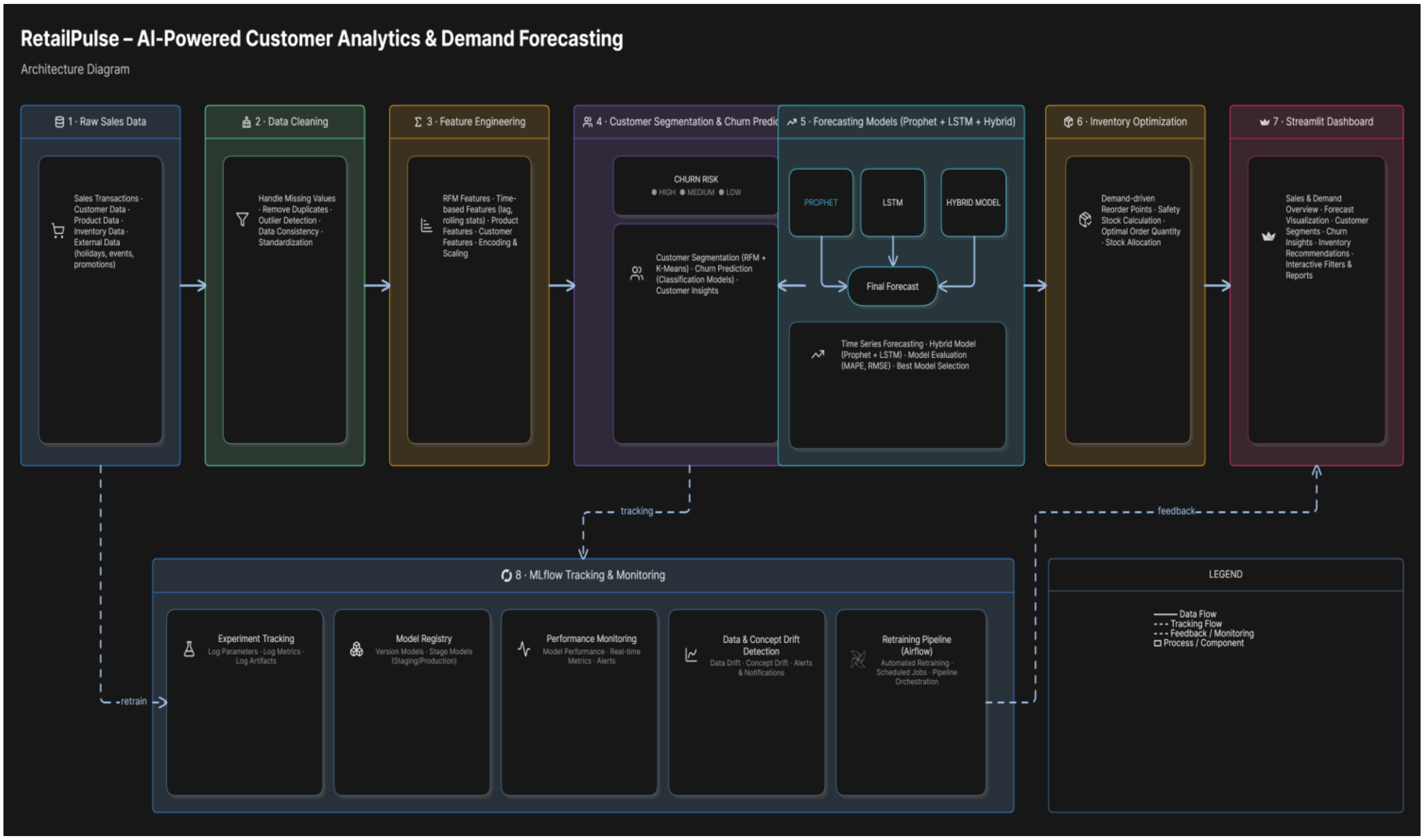
- **Scalable ML pipeline**
- **Low latency dashboard visualization**
- **Reliable model tracking and monitoring**
- **Maintainable and modular architecture**

Architecture & MLOps Workflow

RetailPulse follows an end-to-end AI and MLOps workflow:

Raw Sales Data → Data Cleaning → Feature Engineering → Customer Segmentation & Churn Prediction → Forecasting Models (Prophet + LSTM + Hybrid Forecasting) → Inventory Optimization → Streamlit Dashboard → MLflow Tracking & Monitoring.




RetailPulse – AI-Powered Customer Analytics & Demand Forecasting
Architecture Diagram
1 · Raw Sales Data
Sales Transactions · Customer Data · Product Data · Inventory Data · External Data (holidays, events, promotions)
2 · Data Cleaning
Handle Missing Values · Remove Duplicates · Outlier Detection · Data Consistency · Standardization
3 · Feature Engineering
RFM Features · Time-based Features (lag, rolling stats) · Product Features · Customer Features · Encoding & Scaling
4 · Customer Segmentation & Churn Predic
CHURN RISK
HIGH MEDIUM LOW
Customer Segmentation (RFM + K-Means) · Churn Prediction (Classification Models) · Customer Insights
5 · Forecasting Models (Prophet + LSTM + Hybrid)
PROPHET
LSTM
HYBRID MODEL
Final Forecast
Time Series Forecasting · Hybrid Model (Prophet + LSTM) · Model Evaluation (MAPE, RMSE) · Best Model Selection
6 · Inventory Optimization
Demand-driven Reorder Points · Safety Stock Calculation · Optimal Order Quantity · Stock Allocation
7 · Streamlit Dashboard
Sales & Demand Overview · Forecast Visualization · Customer Segments · Churn Insights · Inventory Recommendations · Interactive Filters & Reports
tracking
feedback
retrain
8 · MLflow Tracking & Monitoring
Experiment Tracking
Log Parameters · Log Metrics · Log Artifacts
Model Registry
Version Models · Stage Models (Staging/Production)
Performance Monitoring
Model Performance · Real-time Metrics · Alerts
Data & Concept Drift Detection
Data Drift · Concept Drift · Alerts & Notifications
Retraining Pipeline (Airflow)
Automated Retraining · Scheduled Jobs · Pipeline Orchestration
LEGEND
Data Flow
Tracking Flow
Feedback / Monitoring
Process / Component

Execution Timeline

Week 1 (15 July - 17 July 2026):

- Requirement gathering.
- Dataset collection, preprocessing.
- Exploratory Data Analysis (EDA)

Week 2 (18 July - 25 July 2026):

- Feature engineering
- Customer segmentation using RFM analysis
- Churn prediction implementation

Week 3 (25 July - 03 Aug 2026):

- Demand forecasting using Prophet and LSTM
- Hybrid model integration
- Inventory optimization

Week 4 (04 Aug - 07 Aug 2026):

- Streamlit dashboard development
- Docker containerization
- Deployment testing
- Final report preparation

Key Features Summary

Feature ID	Feature Name	Description	Acceptance Criteria
F1	Customer Segmentation	Groups customers based on behavior	Customers categorized successfully
F2	Churn Prediction	Predicts at-risk customers	Accurate churn insights generated
F3	Demand Forecasting	Predicts future demand	Reliable forecast trends shown
F4	Hybrid Forecasting	Combines multiple models	Improved prediction quality
F5	Inventory Optimization	Optimizes stock levels	Reduced overstocking risk
F6	Dashboard Analytics	Interactive visualization	Real-time insights accessible
F7	Drift Detection	Detects data changes	Alerts for model degradation
F8	MLflow Tracking	Tracks experiments	Performance history maintained

Technology Stack

Category	Technology	Rationale / Alternatives
Programming	Python	Primary language for ML and analytics
Frontend	Streamlit	Interactive dashboard development
Machine Learning	Scikit-learn, TensorFlow	Model training and prediction
Forecasting	Prophet, LSTM	Time-series forecasting
MLOps	MLflow, Optuna, Airflow	Experiment tracking and orchestration
Containerization	Docker	Portable deployment
Version Control	Git & GitHub	Collaboration and code management

Personal Reflection & Conclusion

Overview: This project provided practical exposure to real-world Machine Learning and MLOps concepts. During development, we learned how to design scalable data pipelines, build predictive models, and deploy applications professionally.

• Hands-on experience with ML workflows.

• End-to-end deployment process using Docker and Streamlit Cloud.

• Practical implementation of time-series forecasting and churn analysis.